Perceived Agency and the Demand for the State: Institutional Legacies in Reunified Germany

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Abstract

This paper studies whether East–West German institutional histories are associated with persistent differences in perceived agency and preferences for state responsibility. The motivation is that the East–West divide remains politically relevant, while locus of control (LOC) is known to matter for economic behavior and may itself be shaped by institutions. Using SOEP data, we first estimate East–West differences in adult LOC by age cohort in 1989 and across time. We then estimate state-responsibility preferences and link prior LOC to later state preferences as a descriptive bridge. East-socialized respondents report more external LOC and stronger pro-state preferences than respondents who were both born in West-Germany and lived there in 1989. Prior more external LOC predicts stronger later pro-state preferences, but adding it only modestly attenuates the East–West preference gap. The same pattern appears for secondary political preference outcomes in the appendix. We therefore interpret LOC as a meaningful psychological correlate of institutional legacies, not as a causally identified mediator or as a sufficient explanation for East–West differences in state preferences and related political attitudes.

1 Introduction

More than three decades after reunification, the East–West divide remains one of the most visible social and political cleavages in Germany. Economic gaps have narrowed, but regional differences in political attitudes, trust, party support, and views about the role of the state continue to shape German public debate [Becker et al., 2020, Bondar and Fuchs-Schündeln, 2023, Pesthy et al., 2021]. Understanding this divide therefore requires more than comparing material outcomes. It also requires studying the beliefs through which individuals interpret their own agency and the responsibilities of the state.

This paper focuses on locus of control (LOC), a belief about whether life outcomes mainly depend on one's own actions or on external forces such as luck, circumstances, institutions, or others. LOC is not only a psychological concept. A large economics literature shows that it is related to labor-market behavior, human-capital investment, job search, and other life outcomes [Cobb-Clark, 2015, Caliendo et al., 2015, Preuss and Hennecke, 2018]. If institutions shape how people experience the link between effort and outcomes, then LOC may also be one channel through which historical political-economic systems leave long-run traces in social and political attitudes.

The division and reunification of Germany offer a useful setting to study this issue. For more than four decades, East and West Germans lived under sharply different political-economic systems. The German Democratic Republic (GDR) organized economic life around centralized planning, extensive state ownership, and broad public responsibility for employment and social protection. The Federal Republic of Germany combined market allocation with a social-insurance welfare state. Reunification placed both populations under a common formal institutional framework, but beliefs and preferences formed under the previous systems may have persisted.

Two literatures motivate our analysis. On the preference side, Alesina and Fuchs-Schündeln [2007] show that East Germans remained more supportive of state intervention after reunification, especially among cohorts with longer exposure to socialism. Bondar and Fuchs-Schündeln [2023] extend this evidence to 2017 and show that East Germans still report stronger pro-state preferences 27 years after reunification, with slower convergence than the original linear extrapolation suggested. On the personality side, Friehe et al. [2015] show that former GDR residents differ in personality measures, including LOC. These studies examine closely related but usually separate outcomes. Alesina & Fuchs-Schündeln and Bondar & Fuchs-Schündeln focus on what individuals think the state should do; Friehe, Pannenberg, and Wedow focus on how individuals perceive their own agency. This paper connects the two by asking whether LOC is a psychological correlate linking institutional exposure and preferences for state responsibility.

We use the German Socio-Economic Panel (SOEP) to study three questions. First, are individuals born in East Germany and living in East Germany in 1989 more likely to report a more external LOC than comparable West-born, West-1989 respondents? Second, does this association vary by life stage at reunification, survey wave, and current regional embeddedness? Third, is prior more external LOC associated with stronger later preferences for state responsibility, and does it account for part of the East–West preference gap documented in the literature?

Our results support a nuanced version of the institutional-legacy hypothesis. East-socialized respondents report a significantly more external LOC than the West reference

group. The gap is about 0.16 standard deviations in a demographic specification and about 0.09 standard deviations after adding socioeconomic controls. The gap is larger among respondents who were adults at reunification, especially those aged 31–45 in 1989, who experienced both adult life under GDR institutions and the disruptive transition after reunification. Across survey waves, the gap is positive in each observed LOC wave, but its size changes over time; composition-weighted wave models and wave-by-cohort checks show that interpretation should account for changing cohort composition.

We also find that East-socialized respondents are more supportive of state responsibility in the exact battery used in Alesina and Fuchs-Schündeln [2007]. Prior more external LOC is positively associated with later pro-state preferences, including when 1999 LOC is linked to 2002 preferences and later LOC is linked to the 2017 repeat battery. However, adding prior LOC only modestly reduces the East–West preference coefficient across different control sets. The same broad lesson appears for the secondary political outcomes in the appendix: LOC is predictive of political interest and satisfaction with democracy, but it does not close the East–West gaps in those outcomes either. We therefore do not interpret LOC as a strong mechanism linking GDR exposure to political preferences. Rather, we interpret it as a theoretically meaningful psychological correlate: East–West institutional histories are associated with both perceived agency and demand for state responsibility, but these attitude gaps reflect a broader bundle of historical, economic, regional, and political channels.

The contribution of the paper is threefold. First, we provide updated evidence on the East–West LOC gap using five SOEP LOC waves from 1999 to 2020, while explicitly treating the 1999 LOC battery as a rescaled earlier instrument. Second, we use the exact Alesina-style preference battery until the last round in 2017 and add a broader state-responsibility measure from the 2021 wave as robustness, building on rather than replacing the update by Bondar and Fuchs-Schündeln [2023]. Third, we show that perceived agency is politically meaningful, but not politically sufficient: it predicts later pro-state preferences, while leaving most of the East–West state-preference gap unexplained. This motivates further work on the party-political expression of these attitudes, for example an exploratory AfD switching analysis described in Appendix C.1.

2 Literature Review

This paper relates to four strands of literature: institutional legacies in reunified Germany, the economic relevance and stability of LOC, the institutional formation of personality and control beliefs, and the role of agency beliefs in demand for redistribution and state intervention.

The first strand studies how political and economic institutions affect preferences even after the institutions themselves change. The German division has become a central setting because East and West Germans shared a common national history before division but experienced sharply different systems after 1949. Alesina and Fuchs-Schündeln [2007] show that East Germans remained more supportive of state intervention after reunification, especially among cohorts with longer exposure to socialism. Bondar and Fuchs-Schündeln [2023] revisit this design with the 2017 SOEP wave and show that the East–West preference gap persists three decades after reunification, while convergence slowed substantially after 2002. This evidence fits into a broader literature on cultural and institutional persistence, which argues that past institutions can leave durable effects on beliefs, norms, and political attitudes even after formal rules converge [Guiso et al., 2006, Nunn, 2012]. At the same time, the German setting should not be treated as a clean randomized experiment: pre-existing regional differences, selective migration, and post-reunification shocks may also matter [Becker et al., 2020]. This is especially relevant for our life-stage interpretation, because cohorts who were adults in 1989 experienced not only longer exposure to GDR institutions, but also the labor-market disruption, occupational uncertainty, and status loss associated with the transition after reunification. We therefore treat East–West exposure as historically informative, but not as random assignment to socialism, and interpret cohort patterns as reflecting both institutional socialization and transition experience.

The second strand examines LOC as an economically relevant belief. LOC captures whether individuals perceive life outcomes as mainly determined by their own actions or by external forces such as luck, fate, circumstances, or powerful others [Rotter, 1966]. A more internal LOC has been linked to labor-market behavior, job search, education, wages, health, and broader economic decision-making [Cobb-Clark, 2015, Caliendo et al., 2015]. In the German labor-market context, Caliendo et al. [2015] show that individuals with a more internal LOC search more intensively for jobs, while individuals with more external beliefs have lower reservation wages. This literature motivates our focus on perceived agency as an economically meaningful outcome rather than a purely psychological side variable.

A third related literature asks whether LOC is stable or shaped by experience. LOC is often treated as relatively stable in adulthood, but not as immutable. Cobb-Clark and Schurer [2013] emphasize substantial persistence, while also documenting changes over the life course; Preuss and Hennecke [2018] show that unemployment temporarily reduces perceived control in the SOEP, even though plant-closure job loss has no long-lasting effect on LOC. This matters for our interpretation. We do not treat LOC as a fixed trait observed after reunification without measurement concerns. Instead, we study whether historical exposure is associated with later adult control beliefs, and we

include socioeconomic controls because current employment, income, education, health, and partnership status may capture contemporaneous circumstances that affect measured LOC.

The possibility that institutions shape control beliefs is supported by work on socialist regimes and personality. Friehe et al. [2015] find that former GDR residents differ from former FRG residents in several personality measures, including LOC. Fuchs-Schündeln and Masella [2016] show that socialist education in East Germany had long-lasting effects on later outcomes, consistent with the broader idea that institutions may shape values and beliefs during formative years. This motivates our life-stage analysis: if institutional experiences matter for LOC, gaps should be larger for cohorts who experienced adult economic and social life under the GDR.

The fourth strand links beliefs about agency, effort, and fairness to preferences for redistribution and state intervention. Models of redistributive preferences often emphasize that individuals support less redistribution when they believe economic outcomes mainly reflect effort, and more redistribution when they believe outcomes are shaped by luck, circumstances, or structural constraints. Fong [2001] shows that beliefs about self-determination are strongly associated with redistributive preferences. Alesina and Angeletos [2005] and Bénabou and Tirole [2006] develop models in which beliefs about effort, fairness, and responsibility sustain different levels of redistribution. More directly for our bridge, Kouba and Pitlik [2014] show that individuals with a more internal LOC and higher self-efficacy report less support for income equalization and government intervention. Our contribution is to bring this agency-and-welfare-state logic into the East–West German institutional-legacy setting.

Overall, the existing literature suggests a clear but still underexplored connection. First, East German exposure predicts stronger preferences for state responsibility and East German exposure predicts a more external LOC. The welfare-state attitude literature predicts that external control beliefs should be associated with stronger demand for state intervention. We study these links jointly in one SOEP design. The central question is not whether LOC is the only mechanism. Preferences for state responsibility may also reflect economic self-interest, regional labor-market conditions, social-insurance experiences, political identity, immigration attitudes, and post-reunification disappointment. Our more modest question is whether prior external LOC is associated with later pro-state preferences and whether including it meaningfully reduces the East–West preference gap.

3 Theoretical Framework

Institutions shape not only the incentives individuals face, but also the way individuals understand the relationship between effort, outcomes, and responsibility. Political-economic systems differ in the extent to which life outcomes are experienced as depending on personal choice, market competition, bureaucratic allocation, or public provision. If repeated institutional experiences shape generalized beliefs about the link between action and outcomes, they may influence both perceived agency and the assignment of responsibility between individuals and the state. This framework builds on the evidence reviewed above that East German exposure is associated with stronger preferences for state responsibility and with differences in LOC. Building on the LOC literature and on evidence about East–West institutional differences, we derive four predictions.

Prediction 1. Respondents born in East Germany and living in East Germany in 1989 should report a more external LOC than comparable respondents born and living in West Germany.

The strength of this association should depend on the length as well as the timing of exposure. Institutional experiences are likely to matter most when they coincide with formative stages of economic and social life, such as schooling, labor-market entry, family formation, and adult interaction with state institutions. Individuals who were adults before reunification had direct experience navigating the GDR system, while those who were children in 1989 had more of their adult life shaped by post-reunification institutions.

It is important to note that adult East Germans in 1989 were also the cohorts most exposed to the economic and institutional disruption of reunification. The transition to a market economy involved labor-market uncertainty, changes in occupational status, the restructuring of firms and local economies, and a sudden change in the rules linking individual effort to economic outcomes. These experiences may themselves have weakened perceived control. For this reason, larger LOC gaps among adult cohorts should be interpreted as evidence of institutional and transition legacies jointly, rather than as a pure effect of GDR socialization.

This life-stage distinction is also consistent with evidence that LOC is persistent but not immutable: perceived control changes over the life course, but adult LOC displays substantial stability. If institutional experiences become embedded in relatively stable adult control beliefs, exposure during labor-market entry, and early adult life should be especially relevant.

Prediction 2. The East–West gap in external LOC should be larger among cohorts that experienced adult life under the GDR and the transition after reunification than among cohorts that were young at reunification.

Individuals who believe that outcomes are strongly shaped by external forces may be more likely to support public responsibility for economic security. This does not require external LOC to imply unconditional support for all forms of state intervention. Rather, it suggests that perceived agency should be related to how individuals divide responsibility between private actors and public institutions in domains of economic security.

Prediction 3. Individuals with a more external LOC should express stronger preferences for state responsibility.

Combining the first three predictions, LOC could potentially explain parts of the East–West preference gap. Individuals socialized in, or strongly affected by, institutional environments where outcomes were experienced as externally shaped may be more likely to assign responsibility for economic security to the state. At the same time, this bridge should not be interpreted as causal mediation: LOC, state preferences, and East–West exposure may all reflect broader historical, socioeconomic, and regional experiences.

Prediction 4. Including LOC in preference regressions should account for part of the persistent East–West preference gap.

In sum, the framework links historical exposure, perceived agency, and demand for state responsibility, but it does not imply a narrow causal mediation story. The empirical analysis therefore tests whether the patterns implied by the agency mechanism are present in the data: whether East-socialized respondents report more external LOC, whether this gap is larger among cohorts exposed during adult life and the reunification transition, whether prior external LOC predicts later state-responsibility preferences, and whether prior LOC helps account for the persistence of the East–West preference gap.

4 Data and Measurement

We use the German Socio-Economic Panel (SOEP), a longitudinal household survey containing information on respondents' demographic characteristics, socioeconomic status, place of birth, residence around reunification, LOC, and political-economic preferences. Because LOC and state-responsibility preferences are observed in different survey years, the analysis does not rely on a single balanced panel. Instead, we use outcome-specific samples for the LOC analysis, the state-preference analysis, and the descriptive prior-LOC-to-later-preference bridge.

Table 1: Main analysis samples

Analysis sample	Outcome	Years	Persons used	Key exposure groups
Main LOC sample	Locus of control	1999, 2005, 2010, 2015, 2020	18,701	West/West vs East/East
Alesina-style preferences	Pro-state preference index	1997, 2002, 2017	15,254	West/West vs East/East
State responsibility	State-responsibility index	2021	8,278	West/West vs East/East
LOC-preference bridge	Prior LOC and later state preferences	Linked waves	10,939–8,000	Descriptive timing link

Notes: Persons used are unique respondents with a valid formative exposure group: West/West, East/East, East/West, or West/East. The bridge counts refer to the exact Alesina-style bridge and the 2021 bridge, respectively. Regression samples can be slightly smaller when controls or outcomes are missing.

Our main exposure variable is based on place of birth and place of residence in 1989. The core East-socialized group consists of respondents born in East Germany and living in East Germany in 1989. The reference group consists of respondents born in West Germany and living in West Germany in 1989. We refer to these groups as East/East and West/West, respectively. All main East-socialization specifications restrict the historical comparison to respondents born by 1989. Additional coefficients identify East-born respondents living in West Germany in 1989 and West-born respondents living in East Germany in 1989, but these crossover groups are smaller and not the main interpretation. We provide the sample diagnostics for these groups in Appendix table 7

Table 2: Exposure-group composition of the main analysis samples

Exposure group	Definition	Main LOC sample	Alesina-style preferences	2021 state responsibility
West/West	Born West; lived West in 1989	12,354	9,977	5,563
East/East	Born East; lived East in 1989	5,399	4,496	2,303
East/West	Born East; lived West in 1989	813	686	370
West/East	Born West; lived East in 1989	135	95	42
Total used	Valid formative exposure group	18,701	15,254	8,278

Notes: Cells report unique persons with valid birthplace and 1989 residence information. West/West is the reference group; East/East is the core East-socialized group. Crossover groups are included in regressions but are not the main comparison.

The main belief outcome is adult LOC. This should not be read as an external-only subscale. The adult LOC battery contains three internal items and seven external items. Our main full LOC index reverse-codes the internal items, combines them with the external items, requires at least five valid items, and standardizes the resulting index within LOC wave. Higher values indicate a more external LOC. The external-only subscale and the internal subscale are used as robustness outcomes. The SOEP LOC scale is measured in the years 1999, 2005, 2010, 2015, and 2020, and the items can be combined into an overall scale with good reliability [Entringer et al., 2022, Specht et al., 2013]. The 1999 LOC battery differs from later waves because it uses a four-point response scale [Preuss and Hennecke, 2018]. We therefore rescale it to the later agreement direction, standardize LOC within wave, include LOC wave fixed effects, and report a no-1999 robustness check.

The main preference outcome follows the state-responsibility battery used by Alesina and Fuchs-Schündeln [2007]. Respondents are asked whether the state or private actors should be responsible for financial security in five domains: family, unemployment, illness, old age, and requiring care. The exact Alesina-style battery is observed in 1997, 2002, and 2017.

We also use a broader state-responsibility battery from 2021 as a secondary extension. Because the 2021 items are not identical to the exact Alesina-style battery, the 2021 outcome is not a direct replication but provides a robustness check; we report it visually in the main results and tabulate the coefficients in the appendix.

Two limitations are important. First, LOC and state preferences are not always measured in the same years. The bridge analysis therefore links prior LOC to later state preferences, especially 1999 LOC to 2002 preferences and 2015 LOC to 2017 preferences. Second, detailed Kreis-level controls are not available to us. The current analysis relies on state-year controls, current-state fixed effects, and current-East diagnostics. A future extension with geocoded SOEP access could add Kreis-level labor-market, industrial-structure, migration, demographic, and public-finance controls; this would allow more granular adjustment for local factors affecting LOC and state preferences and would reduce reliance on a current-East dummy that is difficult to interpret. We describe this possible extension in Appendix C.2.

5 Empirical Strategy

Our empirical strategy documents conditional associations between historical East German exposure, LOC, and preferences for state responsibility. The preferred comparison is between respondents born in East Germany and living in East Germany in 1989 and respondents born in West Germany and living in West Germany in 1989, conditional on

observed covariates. We do not interpret this comparison as random assignment to political systems. Pre-division differences, selective migration, post-reunification economic change, and current regional conditions may all contribute to observed gaps [Becker et al., 2020].

We first estimate the association between East-socialized status and external LOC:

$$ExternalLOC_{it} = \alpha + \beta EastEast_i + G_i'\pi + X_{it}'\gamma + \lambda_t + \varepsilon_{it}, \tag{1}$$

where $ExternalLOC_{it}$ is the wave-standardized full LOC index in the external direction for individual i in LOC wave t, $EastEast_i$ is an indicator for being born in East Germany and living in East Germany in 1989, G_i contains the other non-reference exposure-group indicators, X_{it} denotes controls, and λ_t are LOC-wave fixed effects. Standard errors are clustered at the individual level.

The demographic control block includes gender, age, age squared, and wave fixed effects. The socioeconomic block adds log post-government household income, years of education, unemployment, employment, marital or partnered status, bad health, and migration background. These variables are included because they are important correlates of measured LOC and state preferences, and because labor-market and health shocks can affect stated control beliefs [Preuss and Hennecke, 2018]. They should not be interpreted as neutral pre-treatment controls: some may partly reflect the consequences of reunification and historical exposure. We therefore present demographic and socioeconomic specifications side by side. State-year controls add state unemployment, state employment, and state mean household income. Current-state fixed effects and the current-East dummy are conservative checks, not preferred main specifications, because current residence is highly correlated with historical East exposure and may absorb part of the pathway through which historical exposure matters.

We then examine heterogeneity by life stage at reunification:

$$\begin{aligned} ExternalLOC_{it} =& \alpha + \sum_c \beta_c (EastEast_i \times Cohort1989_{ic}) + \sum_c \delta_c Cohort1989_{ic} \\ &+ G_i'\pi + X_{it}'\gamma + \lambda_t + \varepsilon_{it}. \end{aligned} \tag{2}$$

The fine age-at-1989 bins are 0–9, 10–17, 18–30, 31–45, 46–60, and 61+. The 61+ group is retained in descriptive counts, but is dropped from pooled cohort-effect and wave-by-cohort regressions when it becomes too thin. We also estimate wave-specific models for 1999, 2005, 2010, 2015, and 2020. Because the age composition of East/East and West/West respondents changes across waves, we treat composition-weighted wave models as part of the main evidence: the weighted LOC wave model holds system-group by fine-cohort composition fixed to the 2005 distribution.

The second part of the analysis studies preferences for state responsibility:

$$StatePref_{it} = \alpha + \beta EastEast_i + G_i'\pi + X_{it}'\gamma + \lambda_t + \varepsilon_{it}. \tag{3}$$

The main outcome is the exact Alesina-style pro-state index in 1997, 2002, and 2017. The main pooled models include preference-wave fixed effects. We additionally estimate a 2017-only model and a composition-weighted wave model that fixes system-group by age-at-1989-stage composition to 1997 shares.

Finally, we examine whether prior LOC is associated with later preferences for state responsibility:

$$StatePref_{it} = \alpha + \beta EastEast_i + \theta PriorExternalLOC_{i,t^-} + G_i'\pi + X_{it}'\gamma + \lambda_t + \mu_{t^-} + \varepsilon_{it}. \tag{4}$$

Here $PriorExternalLOC_{i,t^-}$ is the latest prior wave-standardized full LOC index scored in the external direction. The bridge models include preference-year fixed effects and LOC-year fixed effects. The main bridge table reports stepwise specifications with demographics only, socioeconomic controls, and state-year controls. Current-state fixed effects, 2021 bridge estimates, and secondary political outcomes are appendix diagnostics. Comparing the East/East coefficient before and after adding prior LOC is a descriptive attenuation exercise. It does not identify causal mediation, because LOC, preferences, and East–West exposure may all reflect broader historical and socioeconomic experiences.

6 Results

6.1 East–West Differences in Locus of Control

Table 3 shows the central LOC result. East-socialized respondents report a more external LOC than the West-born/West-1989 reference group. In the demographic specification, the East/East coefficient is 0.159 standard deviations. The estimate falls to 0.091 standard deviations after adding socioeconomic controls, but remains highly statistically significant. Adding state-year controls leaves the coefficient very similar at 0.093 standard deviations.

Table 3: East-Socialized Status and External Locus of Control

Specification	East/East 1989	Observations
Demographics + LOC-wave FE	0.159***	47,364
	(0.014)	
Socioeconomic controls + LOC-wave FE	0.091***	46,065
	(0.013)	
State-year controls	0.093***	46,065
	(0.019)	

Notes: Outcome is the full adult LOC battery scored in the external direction and standardized within wave. The reference group is West/West. Standard errors clustered by person are in parentheses. Socioeconomic controls add income, education, labor-market status, partnership, health, and migration background. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

These are the specifications we treat as the main LOC evidence. More demanding current-residence specifications are not used as headline estimates because current residence may itself be a post-reunification outcome or part of the mechanism. Appendix Table 6 and Appendix Figure 4 show these diagnostics. They indicate that the historical East/East coefficient becomes smaller when current East or current-state fixed effects are added, which is expected given the strong overlap between historical East socialization and current East residence. The appendix also shows that excluding the rescaled 1999 LOC wave leaves the main result intact. The result is also consistent across the two conceptual directions of the LOC battery: Appendix Table 6 shows a positive East/East coefficient for the external-only subscale and a negative coefficient for the internal subscale.

6.2 Life Stage at Reunification

Figure 1 shows that the LOC gap is not uniform across cohorts. The gap is smallest among respondents who were children in 1989 and larger among respondents who had already experienced adult life under the GDR. The strongest implied gap is among those aged 31–45 in 1989. This pattern is consistent with the idea that perceived agency is more strongly shaped when institutions are experienced during education, employment, household formation, and direct interaction with state institutions. At the same time, the cohort pattern should not be interpreted as isolating GDR socialization alone. The adult cohorts in 1989 were also the cohorts most exposed to the labor-market disruption, occupational uncertainty, and status changes associated with the transition after reunification. The life-stage evidence therefore points to a combined institutional-and-transition legacy rather than a pure socialist-socialization effect.

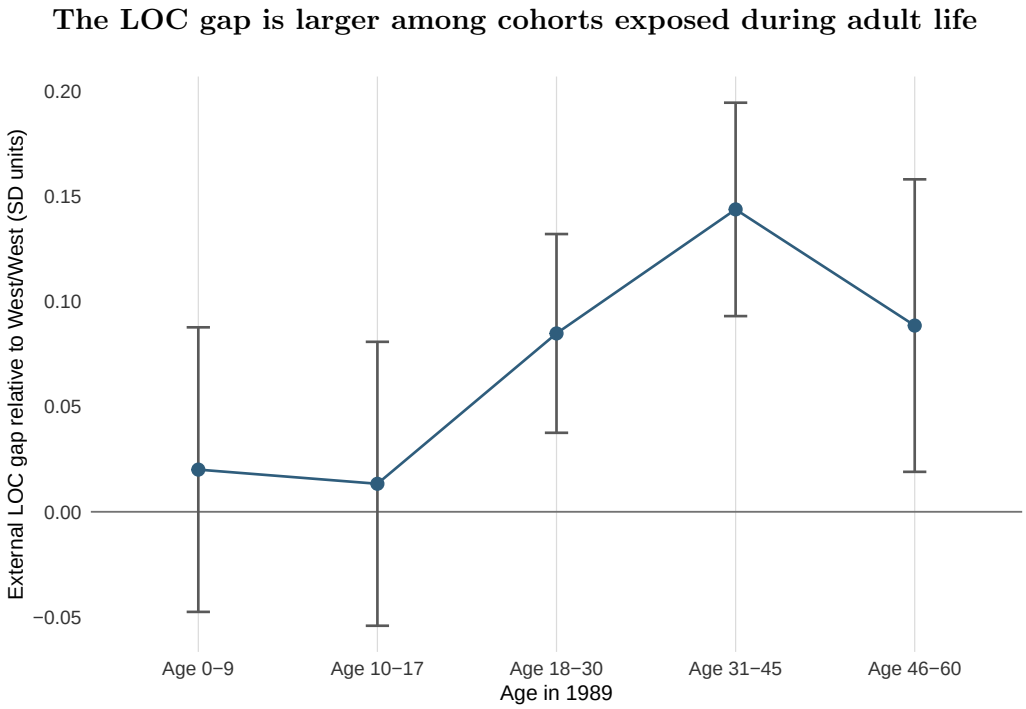


Figure 1: LOC gap by age in 1989. Points show implied East/East external-LOC gaps relative to West/West respondents. Bars are exact 95 percent confidence intervals computed from the clustered variance-covariance matrix of the life-stage interaction model.

Wave-stability evidence is reported in Appendix Figures 5 and 6. The wave evidence is positive across observed LOC waves, including 1999. The gap reduces by 2020 but this is not a clean convergence estimate because the composition of the observed East/East and West/West groups changes across waves. Composition-weighted wave models therefore appear in the appendix as part of the wave evidence. Appendix Table 7 reports the current-residence composition of the LOC sample; we treat these categories descriptively because they combine historical exposure, migration selection, and current regional embeddedness.

6.3 Preferences for State Responsibility

Table 4 shows that East-socialized respondents are more supportive of state responsibility. In the exact Alesina-style index, the East/East coefficient is 0.120 index points in the demographic model and 0.104 after socioeconomic controls. The 2017 exact-repeat wave also shows a positive East/East coefficient. The composition-weighted Alesina wave model fixes system-group by age-at-1989-stage composition to the 1997 distribution and remains positive. We treat this weighted model as part of the main preference evidence because it addresses the same composition problem that matters for interpreting wave comparisons.

Table 4: East-Socialized Status and Preferences for State Responsibility

Specification	East/East coefficient	Observations
Exact Alesina index, demographics + wave FE	0.120***	24,593
	(0.005)	
Exact Alesina index, socioeconomic controls + wave FE	0.104***	23,987
	(0.005)	
Exact Alesina index, 2017 repeat only	0.089***	12,277
	(0.007)	
Exact Alesina index, composition-weighted wave model	0.149***	24,593
	(0.011)	

Notes: Outcome is the exact Alesina-style pro-state index observed in 1997, 2002, and 2017. The 2017 row uses the exact-repeat wave only. The weighted model fixes system-group by age-at-1989 composition to 1997 shares. Standard errors are clustered by person. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

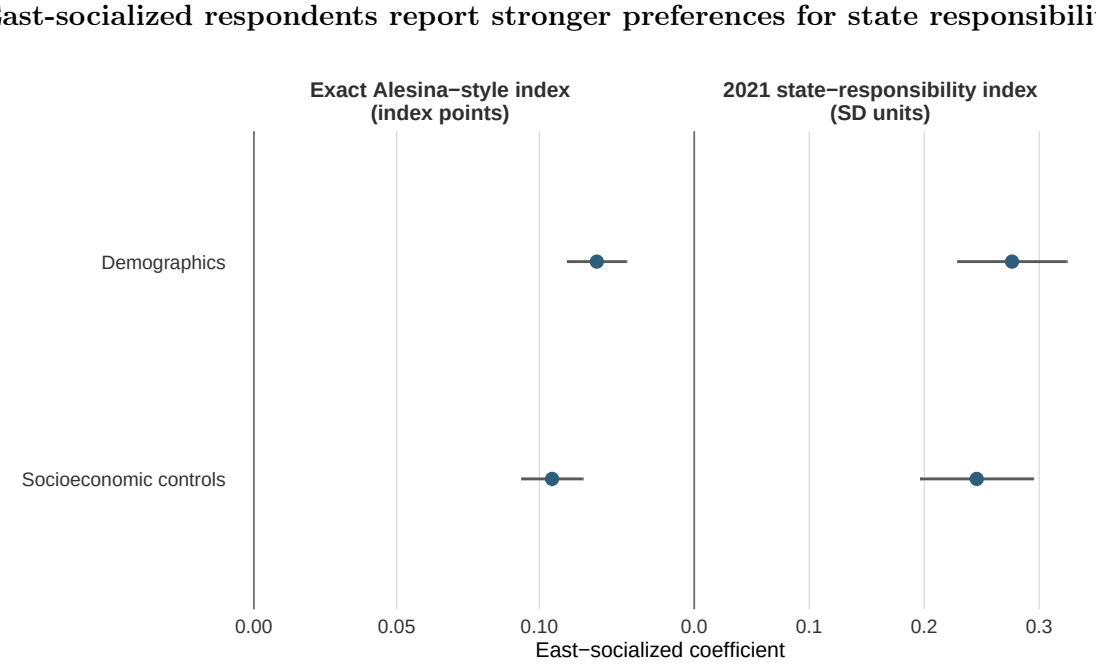


Figure 2: State-responsibility preference gaps. The left panel reports the exact Alesina-style index in index-point units; the right panel reports the broader 2021 state-responsibility index in standard-deviation units. Points show East/East coefficients relative to West/West respondents.

Figure 2 shows that the modern 2021 state-responsibility battery points in the same direction as the exact Alesina-style estimates, although it is not a direct replication because the item wording and scale differ. Appendix Table 8 reports the corresponding 2021 coefficients.

6.4 Prior Locus of Control and Later State Preferences

Table 5 examines whether prior LOC helps bridge the personality and preference results. Prior external LOC is positively associated with later pro-state preferences. In the exact Alesina bridge, the prior LOC coefficient is about 0.116 to 0.143 standard deviations depending on the control set. This is the theoretically expected direction: respondents with more external perceived agency are more supportive of state responsibility in later preference waves.

Table 5: Prior External LOC and Later State Preferences

Bridge specification	Prior LOC	East/East without LOC	East/East with LOC	Attenuation
Exact Alesina, demographics	0.143***	0.309***	0.288***	0.020
	(0.009)	(0.019)	(0.019)	
Exact Alesina, socioeconomic	0.116***	0.267***	0.257***	0.010
	(0.009)	(0.020)	(0.020)	
Exact Alesina, state-year controls	0.116***	0.195***	0.184***	0.011
	(0.009)	(0.032)	(0.031)	

Notes: Prior LOC is the latest prior wave-standardized LOC index scored in the external direction. Models include preference-year and LOC-year fixed effects. The state-preference outcome is standardized, so coefficients are in standard-deviation units. Attenuation is the change in the East/East coefficient after adding prior LOC. This is descriptive, not causal mediation. Standard errors are clustered by person. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Figure 3 shows that in the demographics-only exact Alesina bridge, adding prior LOC reduces the East/East coefficient by about 0.020 standard deviations. In the socioeconomic and state-year specifications, the corresponding reductions are about 0.010 and 0.011. Thus, prior LOC predicts later state preferences, but it does not meaningfully explain away the East/East preference gap. This is one of the central results of the paper. LOC is politically meaningful, but the East–West state-preference gap is not simply an LOC gap. One interpretation is that the remaining gap also captures other parts of the East German historical experience, including post-reunification labor-market disruption, regional embeddedness, political identity, and collective memory, which are not fully summarized by individual LOC. Appendix Table 9 reports the current-state-FE and 2021 bridge checks.

External LOC predicts pro-state preferences, but explains little of the East-West gap

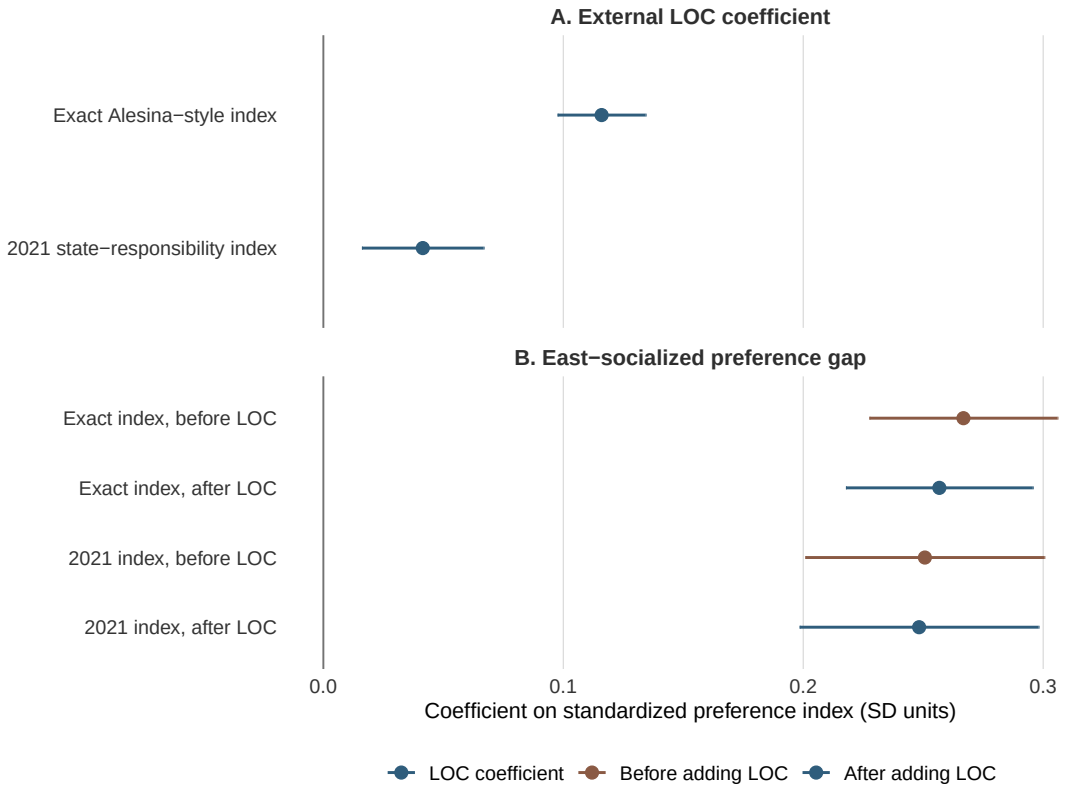


Figure 3: Prior LOC and state-preference gaps. Panel A reports the association between prior external LOC and later pro-state preferences. Panel B compares the East/East preference gap before and after adding prior LOC.

Appendix Table 10 shows the same broad pattern for two secondary political outcomes: political interest and satisfaction with democracy. More external same-year LOC is associated with higher political interest and lower satisfaction with democracy, but adding LOC changes the East/East coefficients only modestly. This reinforces the interpretation that perceived agency is part of a broader political-attitudes profile rather than a mechanism that explains away East–West differences in state preferences. It also motivates the exploratory AfD extension in Appendix C.1, where the next step is to study whether changing party support after 2015 is more closely related to immigration concern, democratic dissatisfaction, prior LOC, pro-state preferences, or current-East embeddedness.

7 Discussion

The results point to a persistent association between East German historical exposure, perceived agency, and demand for state responsibility. East socialized respondents report a more external LOC, and this difference remains visible after demographic and socioeconomic adjustment. The gap is not equally large across cohorts or waves, which matters for interpretation. The cohort pattern suggests that exposure during adult life is more strongly associated with later LOC than exposure only in childhood. However, adult exposure should be understood broadly. Respondents who were adults in 1989 had not only spent more of their formative economic and social lives under GDR institutions; they were also more directly exposed to the uncertainty, labor-market restructuring, occupational status changes, and institutional rupture of the reunification transition. The cohort result should therefore be interpreted as evidence of a broader East German institutional-and-transition legacy, rather than as a clean separation between socialist socialization and post-reunification shock. The wave pattern suggests persistence, but also changing composition and possible attenuation.

The preference results are consistent with the Alesina-style institutional-legacy argument: East socialized respondents remain more supportive of state responsibility, in line with the long-run persistence documented by Bondar and Fuchs-Schündeln [2023]. The bridge results add a psychological-belief dimension. External LOC predicts stronger pro-state preferences, which provides support for the prediction that perceived agency helps account for the persistence of the East–West preference gap. However, the magnitude is more limited than a strong agency-channel interpretation would imply. Adding prior LOC only modestly attenuates the East/East preference coefficient. This suggests that perceived agency is politically meaningful, but it does not explain a significant part of the persistence of East–West differences in state-responsibility preferences. The remaining gap likely reflects other channels, including economic structure, post-reunification labor-market experiences, political identity, collective memory, current regional embeddedness, and immigration attitudes.

The same logic carries over to the secondary political outcomes. External LOC is associated with political interest and satisfaction with democracy, but the East/East coefficients remain large after adding LOC. This is important for the broader motivation of the paper: the East–West differences studied here cannot be reduced to one psychological belief. The appendix therefore treats AfD switching as a natural next step. Existing work shows that immigration attitudes, nativism, and populist orientations are relevant to the East–West gap in AfD support [Lange, 2021, Pesthy et al., 2021]. Our exploratory diagnostics suggest that later AfD support appears among a broader mix of respondents with no

prior party preference and respondents from mainstream parties. Future versions should estimate this political-process extension more systematically.

Several limitations remain. First, the East–West comparison is not random assignment. Pre-division differences, selective migration, reunification shocks, and current regional environments may all affect the estimates. Second, current-East and current-state controls are difficult to interpret because they can absorb post-reunification embeddedness as well as local conditions. Third, we are missing the geocoded SOEP data to include Kreis-level controls to create a more credible design around regional influences. Fourth, the LOC-preferences bridge is descriptive and does not establish a causal mechanism.

Future work could strengthen the design in two directions. The first is a Kreis-level extension using geocoded SOEP access. Granular controls for unemployment, industry structure, income, demographic change, migration, social infrastructure, and public finance would help adjust for local factors that plausibly affect LOC and preferences. This would also allow the analysis to rely less on the blunt current-East dummy. The second is a stronger political-party extension studying whether post-2015 AfD support is correlated with prior dissatisfaction, immigration concerns, LOC, pro-state preferences, and East/current-East exposure. In the appendix we provide an initial assessment of these extensions.

8 Conclusion

This paper connects two important findings in the literature on reunified Germany: East German exposure is associated with stronger preferences for state responsibility, and it is associated with a more external LOC. Using updated SOEP LOC waves and data on political preferences, we show that East-socialized respondents report more external perceived agency and stronger pro-state preferences. The LOC gap varies by life stage at reunification and remains visible across waves.

External LOC is positively associated with pro-state preferences. Yet adding LOC only modestly attenuates the East–West preference gap, and the same qualitative pattern appears for secondary political outcomes in the appendix. The main conclusion is therefore not that LOC is the single causal mechanism. Rather, LOC is a meaningful psychological correlate that helps connect institutional histories, perceived agency, and the demand for the state. The legacy of East Germany does not persist through LOC alone. It likely persists through a mixture of beliefs, post-reunification experiences, political developments, and regional conditions, so no single measure can fully explain the remaining East–West differences.

References

Alberto Alesina and George-Marios Angeletos. Fairness and redistribution. *American Economic Review*, 95(4):960–980, 2005. doi: 10.1257/0002828054825655.

Alberto Alesina and Nicola Fuchs-Schündeln. Good-bye Lenin (or not?): The effect of communism on people's preferences. *American Economic Review*, 97(4):1507–1528, 2007. doi: 10.1257/aer.97.4.1507.

Sascha O. Becker, Lukas Mergele, and Ludger Woessmann. The separation and reunification of Germany: Rethinking a natural experiment interpretation of the enduring effects of communism. *Journal of Economic Perspectives*, 34(2):143–171, 2020. doi: 10.1257/jep.34.2.143.

Roland Bénabou and Jean Tirole. Belief in a just world and redistributive politics. *The Quarterly Journal of Economics*, 121(2):699–746, 2006. doi: 10.1162/qjec.2006.121.2.699.

Mariia Bondar and Nicola Fuchs-Schündeln. Good bye Lenin revisited: East–West preferences three decades after German reunification. *German Economic Review*, 24(1):97–119, 2023. doi: 10.1515/ger-2022-0042.

Marco Caliendo, Deborah A. Cobb-Clark, and Arne Uhlendorff. Locus of control and job search strategies. *The Review of Economics and Statistics*, 97(1):88–103, 2015. doi: 10.1162/REST_a_00459.

Deborah A. Cobb-Clark. Locus of control and the labor market. *IZA Journal of Labor Economics*, 4(1):3, 2015. doi: 10.1186/s40172-014-0017-x.

Deborah A. Cobb-Clark and Stefanie Schurer. Two economists' musings on the stability of locus of control. *The Economic Journal*, 123(570):F358–F400, 2013. doi: 10.1111/ecoj.12069.

Theresa M. Entringer, Florian Griese, Stefan Zimmermann, and David Richter. SOEP scales manual: Updated for SOEP-Core v37. SOEP Survey Papers 1217, DIW Berlin/SOEP, Berlin, 2022. URL https://www.diw.de/documents/publikationen/73/diw_01.c.862242.de/diw_ssp1217.pdf.

Christina Fong. Social preferences, self-interest, and the demand for redistribution. *Journal of Public Economics*, 82(2):225–246, 2001. doi: 10.1016/S0047-2727(00)00141-9.

Tim Friehe, Markus Pannenberg, and Michael Wedow. Let bygones be bygones? socialist regimes and personalities in Germany. CESifo Working Paper Series 5440, CESifo, 2015. URL https://ssrn.com/abstract=2636243.

Nicola Fuchs-Schündeln and Paolo Masella. Long-lasting effects of socialist education. *The Review of Economics and Statistics*, 98(3):428–441, 2016. doi: 10.1162/REST_a_00583.

Luigi Guiso, Paola Sapienza, and Luigi Zingales. Does culture affect economic outcomes? *Journal of Economic Perspectives*, 20(2):23–48, 2006. doi: 10.1257/jep.20.2.23.

Luděk Kouba and Hans Pitlik. I wanna live my life: Locus of control and support for the welfare state. MENDELU Working Papers in Business and Economics 46/2014, Mendel University in Brno, 2014. URL https://ideas.repec.org/p/men/wpaper/46_2014.html.

Martin Lange. The legacy of state socialism on attitudes toward immigration. *Journal of Comparative Economics*, 49(3):733–750, 2021. doi: 10.1016/j.jce.2020.12.005.

Nathan Nunn. Culture and the historical process. *Economic History of Developing Regions*, 27(sup1):S108–S126, 2012. doi: 10.1080/20780389.2012.664864.

Maria Pesthy, Matthias Mader, and Harald Schoen. Why is the AfD so successful in eastern Germany? an analysis of the ideational foundations of the AfD vote in the 2017 federal election. *Politische Vierteljahresschrift*, 62(1):69–91, 2021. doi: 10.1007/s11615-020-00285-9.

Malte Preuss and Juliane Hennecke. Biased by success and failure: How unemployment shapes locus of control. *Labour Economics*, 53:63–74, 2018. doi: 10.1016/j.labeco.2018.05.007.

Julian B. Rotter. Generalized expectancies for internal versus external control of reinforcement. *Psychological Monographs: General and Applied*, 80(1):1–28, 1966. doi: 10.1037/h0092976.

Jule Specht, Boris Egloff, and Stefan C. Schmukle. Everything under control? the effects of age, gender, and education on trajectories of perceived control in a nationally representative German sample. *Developmental Psychology*, 49(2):353–364, 2013. doi: 10.1037/a0028243.

A Appendix: LOC Robustness and Diagnostic Figures

This appendix collects the LOC checks that are important for transparency but too detailed for the main results. The main paper uses the demographic, socioeconomic, and state-year-control specifications as headline estimates. The checks below show how the

estimates behave when current residence is added, when the 1999 LOC wave is excluded, and when alternative LOC subscales are used.

Table 6: LOC Robustness and Current-Residence Diagnostics

Specification	East/East 1989	Current East	Observations
Current-state fixed effects	0.051**	–	46,065
	(0.025)		
Current-East dummy check	0.025	0.077***	46,065
	(0.025)	(0.025)	
No-1999 socioeconomic robustness	0.096***	–	42,113
	(0.014)		
No-1999 state-year robustness	0.110***	–	42,113
	(0.019)		
External-only subscale	0.061***	–	46,053
	(0.013)		
Internal subscale	−0.100***	–	46,053
	(0.014)		

Notes: Reference group is West/West. Current-East models are diagnostic because current residence and East/East exposure are highly correlated (correlation 0.861; VIFs about 4.2). No-1999 rows exclude the rescaled 1999 LOC battery. Standard errors are clustered by person. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

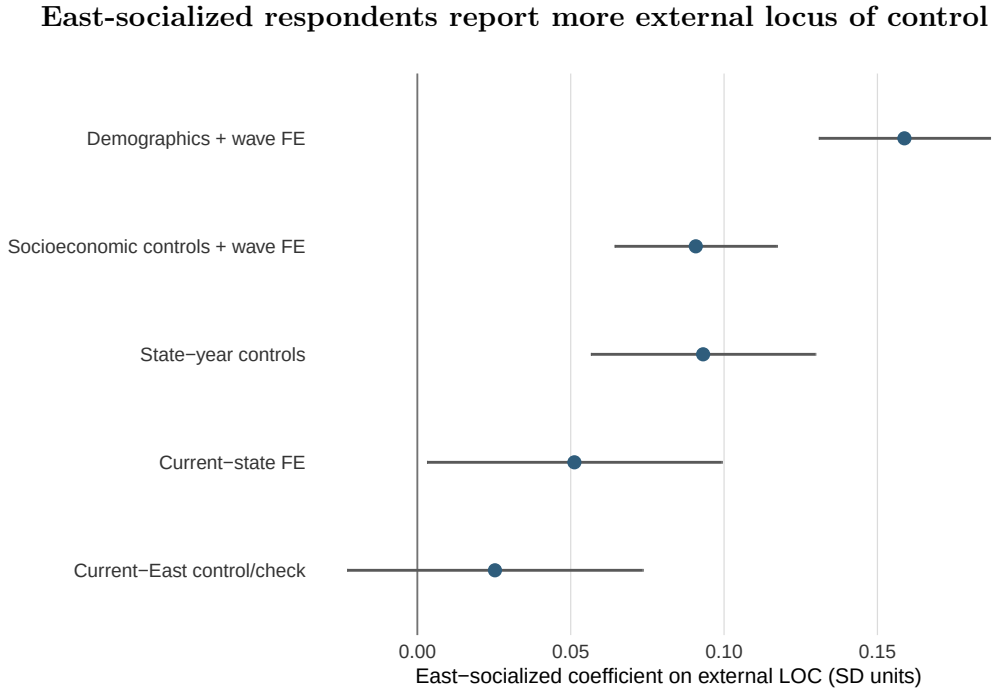


Figure 4: LOC specification checks. Points show East/East coefficients relative to West/West respondents across the main LOC specifications and conservative current-residence diagnostics.

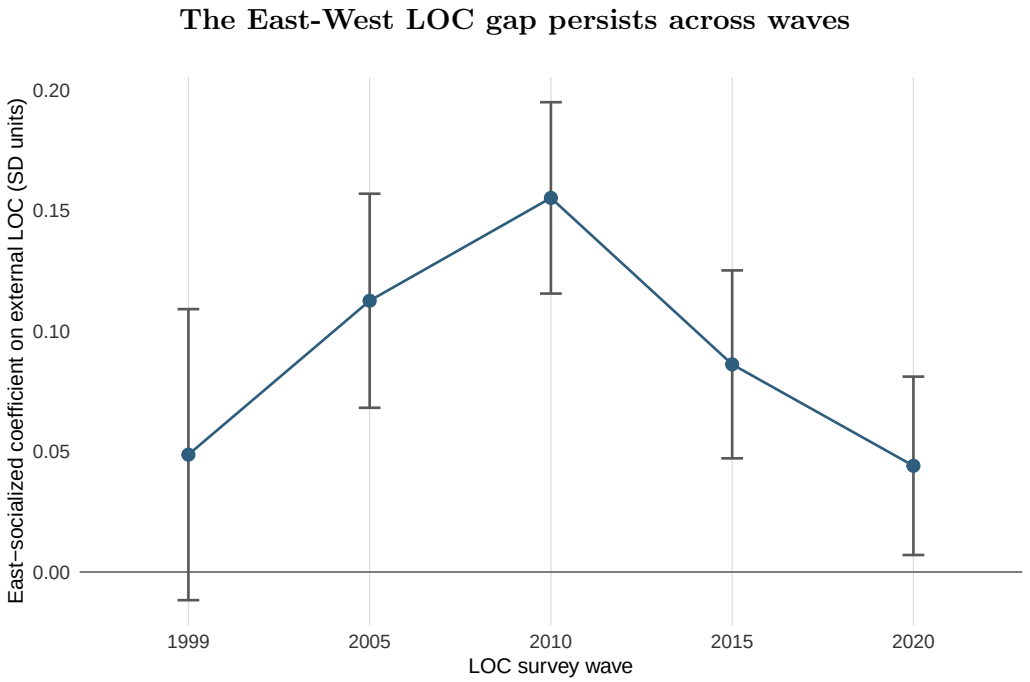


Figure 5: Wave-specific LOC gaps. Points show East/East coefficients relative to West/West respondents in each LOC wave. The 1999 wave uses the rescaled four-point LOC battery.

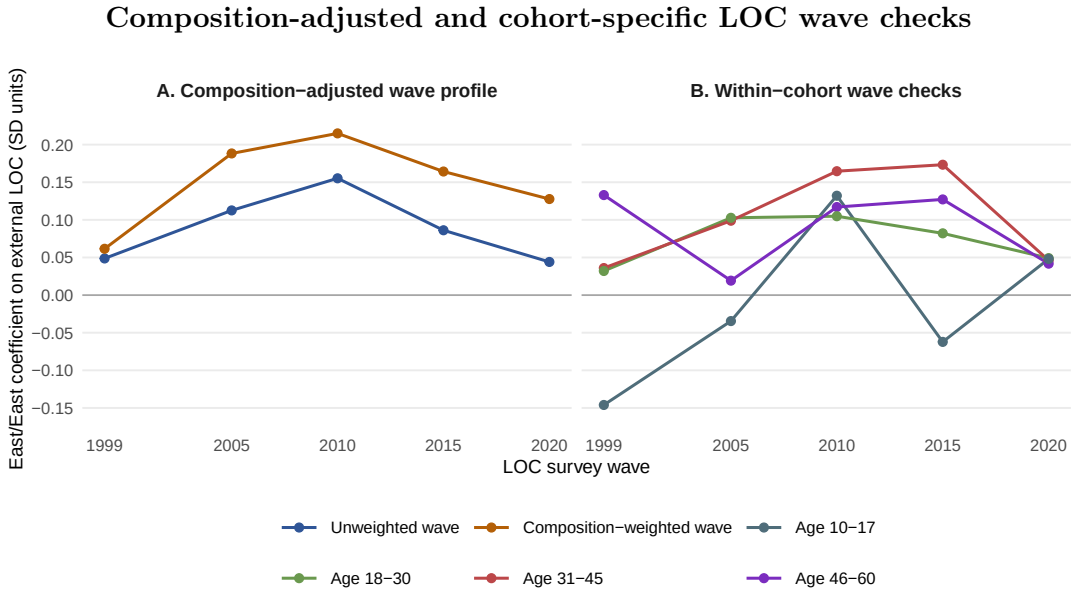


Figure 6: Composition-adjusted and cohort-specific LOC wave checks. Panel A compares unweighted wave-specific East/East coefficients with point estimates from a composition-weighted wave-interaction model that fixes system-group by fine-cohort composition to the 2005 distribution. Panel B plots wave-specific East/East coefficients within fine age-in-1989 cohorts that satisfy the support rule in every LOC wave.

Table 7: Current Residence Composition in the LOC Sample

Formative exposure group	Current West obs. (persons)	Current East obs. (persons)
West/West in 1989	29,948 (12,175)	424 (248)
East/East in 1989	1,943 (883)	12,493 (4,715)
East/West in 1989	2,033 (744)	160 (91)
West/East in 1989	141 (60)	233 (80)

Notes: Cells report LOC person-wave observations, with unique persons in parentheses, from the 1999-inclusive LOC sample. Current residence is measured at the observation wave, so these categories describe observed current embeddedness rather than exact migration-event histories. Respondents observed under more than one current-residence category can contribute to more than one person count.

B Appendix: State Preferences and Prior-LOC Bridge Checks

The main text reports compact tables and readable figures for the exact Alesina-style state-preference index and the prior-LOC bridge. This appendix keeps the non-headline 2021 and current-state checks.

Table 8: 2021 State-Responsibility Extension

Specification	East/East coefficient	Observations
2021 broader state-responsibility index, demographics	0.276^{***}	8,275
	(0.024)	
2021 broader state-responsibility index, socioeconomic controls	0.246^{***}	7,794
	(0.025)	

Notes: The 2021 outcome is standardized and comes from a broader, non-identical state-responsibility battery. Standard errors are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 9: Additional Prior-LOC Bridge Checks

Bridge specification	Prior LOC	East/East without LOC	East/East with LOC	Attenuation
Exact Alesina, current-state FE	0.116***	0.122***	0.116***	0.006
	(0.009)	(0.039)	(0.038)	
2021 state-responsibility extension	0.041***	0.251***	0.248***	0.002
	(0.013)	(0.025)	(0.025)	

Notes: Current-state fixed effects are conservative because current state may absorb regional embeddedness. The 2021 row uses the broader, non-identical state-responsibility battery. Standard errors are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

B.1 Secondary Political Outcomes

Table 10 reports two same-year secondary political outcomes: political interest and satisfaction with democracy. These are not substitutes for the state-responsibility measures, but they show that external LOC is associated with a broader political-attitudes profile. The same qualitative pattern holds as in the main bridge analysis: LOC is predictive, while the East/East coefficients remain large after adding LOC.

Table 10: Secondary political outcomes

Outcome	External LOC	East/East without LOC	East/East with LOC	Observations
Political interest	0.057***	0.097***	0.091***	42,057
	(0.005)	(0.012)	(0.012)	
Satisfaction with democracy	−0.295***	−1.066***	−1.036***	30,581
	(0.017)	(0.034)	(0.034)	

Notes: Models use the same-year LOC-politics sample with socioeconomic controls and survey-wave fixed effects. External LOC is the full adult LOC index scored in the external direction. East/East is relative to West/West. Standard errors are clustered by person. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

C Appendix: Exploratory Extensions

C.1 AfD Party-Preference Extension

The AfD extension is exploratory and descriptive. It is motivated by the increasing party-political salience of the East–West divide and by evidence that immigration at-

titudes, nativism, and populist orientations are important for AfD support in eastern Germany [Lange, 2021, Pesthy et al., 2021]. The extension uses a strict pre/post panel in which party state is observed in 2010 and the first post-2015 party state is observed in 2017, 2019, 2020, or 2021. It asks whether post-2015 AfD support appears to come primarily from Die Linke, from mainstream parties, from other parties, or from respondents with no prior party preference. It also defines a possible future switching model using prior immigration concern, concern about hostility toward foreigners, satisfaction with democracy, political interest, prior LOC, prior or same-year pro-state preferences, and East/current-East exposure. This extension should not be interpreted as a causal voting analysis; its purpose is to show where a more complete political-process analysis should go next.

The preliminary counts show why the extension should not be framed as a simple Die-Linke-to-AfD story. In the strict pre/post sample there are 8,737 matched respondents, 196 later AfD supporters, and only 17 strict Die Linke-to-AfD cases. The larger origin categories are mainstream-party support and no party preference in 2010. Rolling adjacent-transition diagnostics are larger, with 1,131 any-to-AfD transitions and 52 left-to-AfD transitions, but these are event counts rather than the preferred strict pre/post design.

Table 11: Exploratory AfD Extension: Prior 2010 Party State Among Later AfD Supporters

Current region in 2010	Later AfD supporters	Die Linke	Mainstream	No party preference
All respondents	196	17 (8.7%)	73 (37.2%)	84 (42.9%)
Current West	114	8 (7.0%)	51 (44.7%)	48 (42.1%)
Current East	82	9 (11.0%)	22 (26.8%)	36 (43.9%)

Notes: Strict pre/post sample: party state observed in 2010 and first observed post-2015 party state observed in 2017, 2019, 2020, or 2021. Omitted origin category is "other party".

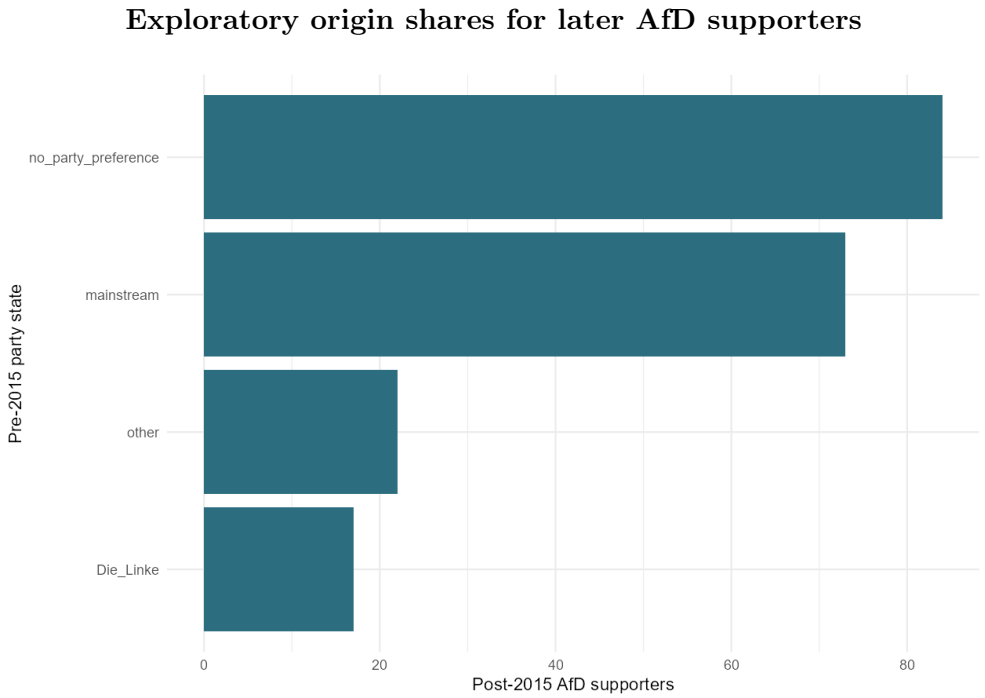


Figure 7: Exploratory origin shares for later AfD supporters in the strict pre/post sample. This figure supports the appendix interpretation that later AfD support is not mainly a Die-Linke-to-AfD transition in the observed strict panel.

C.2 Kreis-Level Controls Extension

A second extension would use Kreis-level controls if geocoded SOEP access becomes available. A prepared external Kreis-year panel already exists in `data_kreis_controls/processed/`, but the current SOEP extract does not contain the protected respondent-to-Kreis link required to merge these controls to individual LOC or preference observations. Therefore we do not estimate Kreis-level models in this paper.

Substantively, this extension would improve the regional-control strategy. Instead of relying on a coarse current-East dummy, the preferred model would add blockwise local controls for labor markets, income and productivity, demographics and migration, urbanicity, education, and sectoral structure. The prepared panel includes variables such as Kreis unemployment, long-term unemployment, SGB II or benefit-recipient rates, disposable income per capita, GDP per capita, gross wages, foreigner share, net migration, population density, age structure, school-leaver qualification shares, and sectoral employment shares. The sources are the Federal Employment Agency unemployment series, INKAR 2025, and regional accounts from VGRDL. Even with these controls, the extension would remain observational because current residence and migration after reunification are themselves endogenous; its value would be more granular adjustment for local conditions, not causal identification by itself.